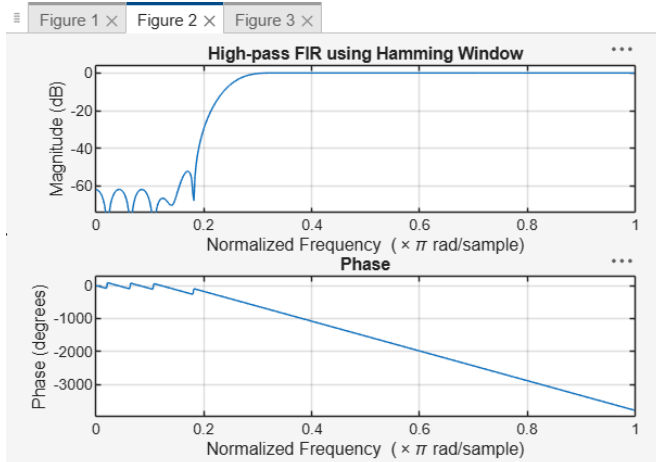
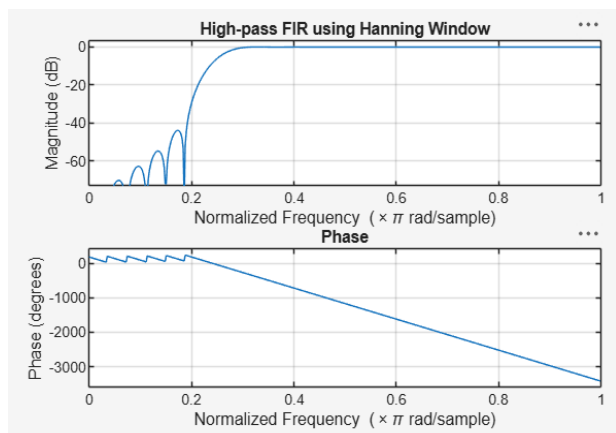
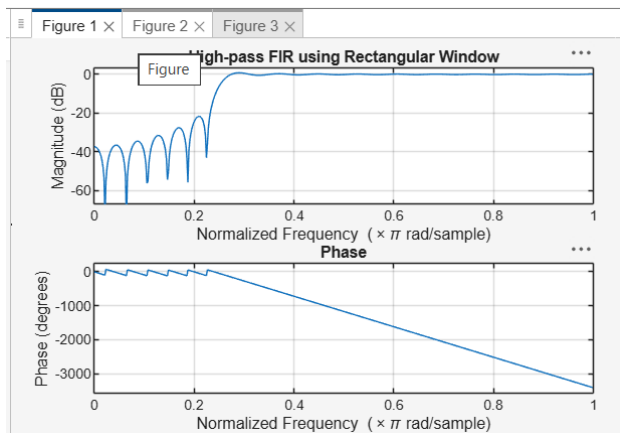


Experiment 5

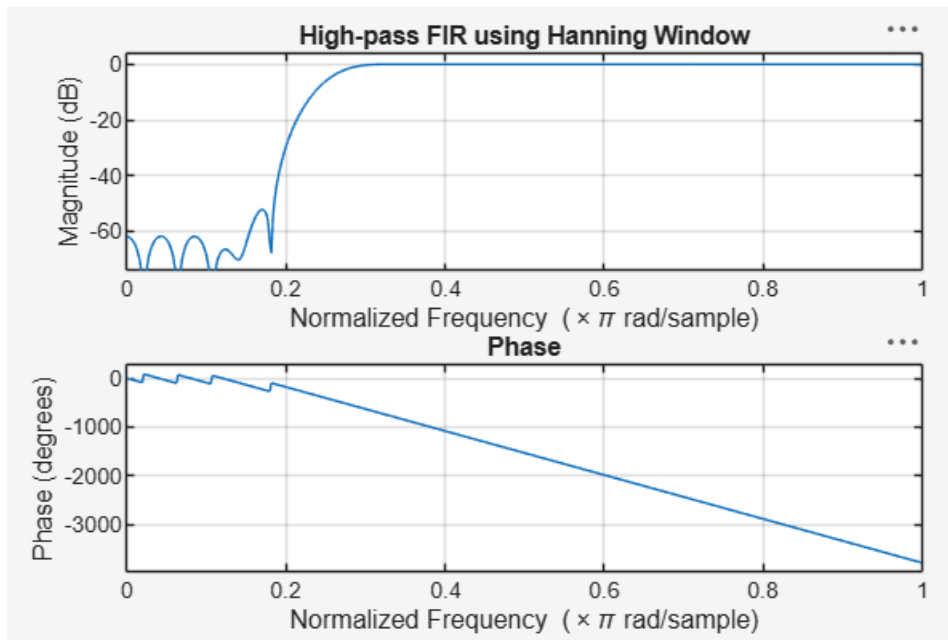
5.1 Generate the coefficients of an FIR High-Pass filter using Rectangular windowing techniques.

```
clc; clear; close all; %% --- Input Parameters --- fp = 1000; % High-pass cutoff frequency (Hz)
fs = 8000; % Sampling frequency (Hz) N = 50; % Filter order wc = fp/(fs/2); % Normalized
cutoff (0-1) %% --- 1. FIR High Pass using Rectangular Window --- b_rect = fir1(N, wc, 'high',
rectwin(N+1)); disp('--- High-Pass Coefficients (Rectangular Window) ---'); disp(b_rect); %%
--- 2. FIR High Pass using Hamming Window --- b_hamm = fir1(N, wc, 'high', hamming(N+1));
disp('--- High-Pass Coefficients (Hamming Window) ---'); disp(b_hamm); %% --- 3. FIR High
Pass using Hanning Window --- b_hann = fir1(N, wc, 'high', hanning(N+1)); disp('--- High-
Pass Coefficients (Hanning Window) ---'); disp(b_hann); %% --- Frequency Response Plots
--- figure; freqz(b_rect,1); title('High-pass FIR using Rectangular Window'); figure;
freqz(b_hamm,1); title('High-pass FIR using Hamming Window'); figure; freqz(b_hann,1);
title('High-pass FIR using Hanning Window');
```



5.2. Generate the coefficients of an FIR High-Pass filter using Hamming windowing techniques

```
clc; clear; close all; %% --- Input Parameters --- fp = 1000; % High-pass cutoff frequency (Hz)
fs = 8000; % Sampling frequency (Hz) N = 50; % Filter order wc = fp/(fs/2); % Normalized
cutoff (0-1) %% --- FIR High Pass using Hamming Window --- b_hamm = fir1(N, wc, 'high',
hamming(N+1)); disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm); %% --- Frequency Response Plots --- figure; freqz(b_hamm,1); title('High-
pass FIR using Hamming Window'); title('High-pass FIR using Hanning Window');
```



5.3. Generate the coefficients of an FIR High-Pass filter using Hanning windowing techniques

```
clc; clear; close all; %% --- Input Parameters --- fp = 1000; % High-pass cutoff frequency (Hz)
fs = 8000; % Sampling frequency (Hz) N = 50; % Filter order wc = fp/(fs/2); % Normalized
cutoff (0–1) %% --- FIR High Pass using Hanning Window --- b_hann = fir1(N, wc, 'high',
hanning(N+1)); disp('--- High-Pass Coefficients (Hanning Window) ---'); disp(b_hann'); %% -
-- Frequency Response Plots --- figure; freqz(b_hann,1); title('High-pass FIR using Hanning
Window');
```

